

Lesson End Project

Designing a Network Solution

Project agenda: To design a network solution

Description: You are working as a principal engineer in an organization, and you have been asked to design a solution for an on-premises network to deploy a virtual appliance. The plan is to deploy multiple Azure VMs and connect the on-premises network to Azure using a site-to-site connection. You need to also ensure that all the network traffic that will be directed from the Azure VMs to a specific subnet must flow through the virtual appliance. As part of compliance, the VMs should not access the internet as they store sensitive data. Recommend a design considering the above requirements.

Tools required: Azure account with administrator credentials

Prerequisites: None

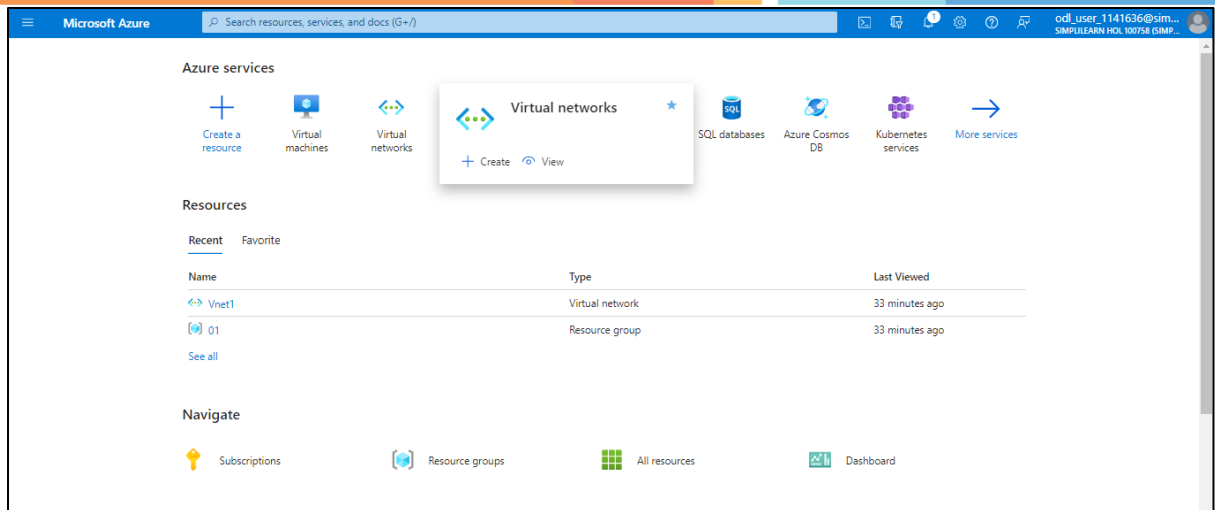
Expected deliveries: Added the screenshots for every step to design a network solution.

Steps to be followed:

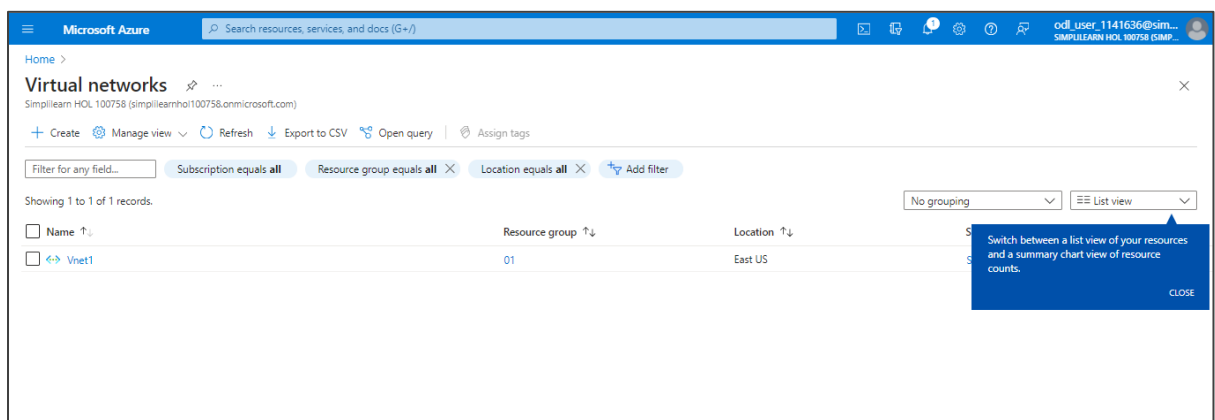
1. Create a virtual network
2. Deploy the virtual machine in the respective virtual network
3. Verify the network security group of the virtual machine

Step 1: Create a virtual network

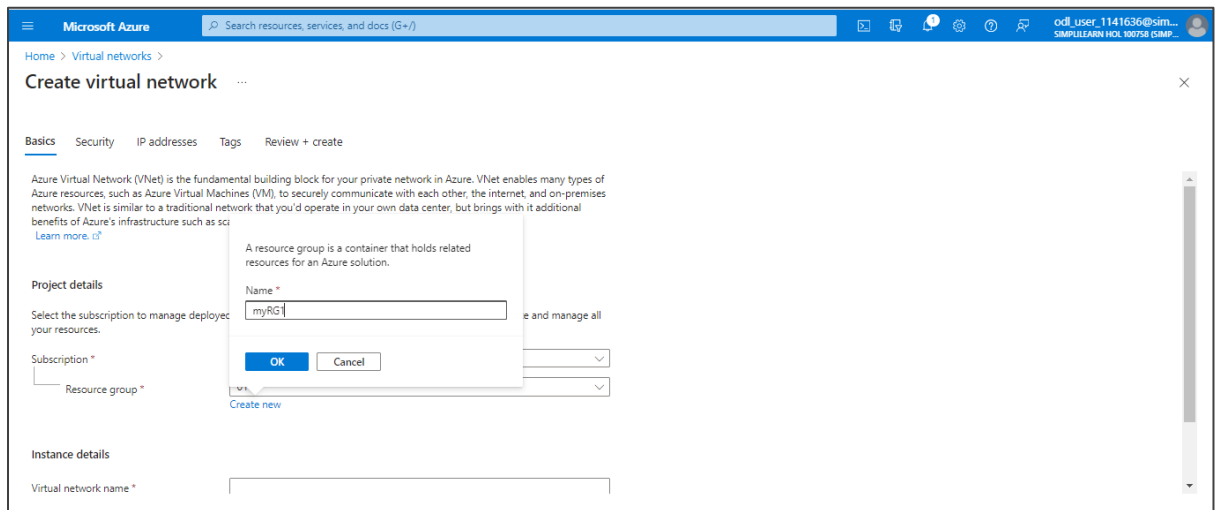
- 1.1 Log in to the Azure portal, search and select **Virtual networks**



2.1 Click on the + Create



2.2 On the **Basics** page, under **Resource group** create a new resource group (**myRG1**)



Microsoft Azure | Search resources, services, and docs (G+)

Home > Virtual networks > Create virtual network

Basics | Security | IP addresses | Tags | Review + create

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scalability and high availability.

[Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *
 Resource group *

A resource group is a container that holds related resources for an Azure solution.

Name *
 myRG1

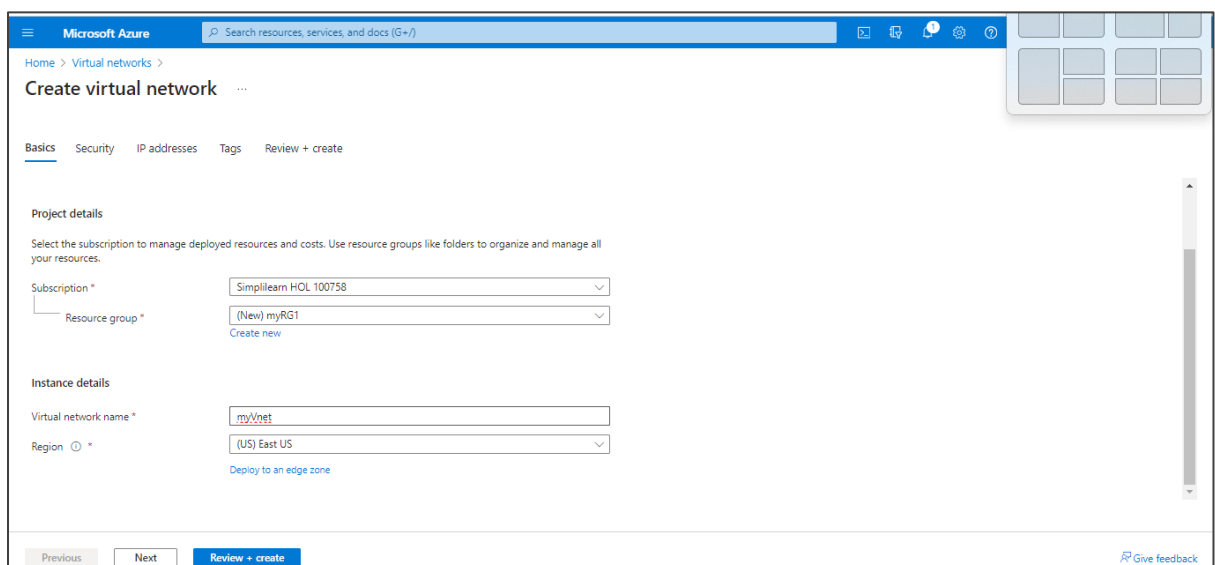
OK Cancel

Create new

Instance details

Virtual network name *

2.3 Under **Instance details**, enter **Virtual network name** as **myVnet** and click on **Review + create**



Microsoft Azure | Search resources, services, and docs (G+)

Home > Virtual networks > Create virtual network

Basics | Security | IP addresses | Tags | Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *
 Resource group *

Simplilearn HOL 100758
 (New) myRG1

Create new

Instance details

Virtual network name *
 myVnet

Region *
 (US) East US

Deploy to an edge zone

Previous Next Review + create

[Give feedback](#)

2.4 Click on the **Create**

The screenshot shows the 'Create virtual network' wizard in the Microsoft Azure portal. The 'Review + create' tab is selected. The form displays the following configuration:

- Basics**
 - Subscription: Simplilearn HOL 100758
 - Resource Group: myRG1
 - Name: myVnet
 - Region: East US
- Security**
 - Azure Bastion: Disabled
 - Azure Firewall: Disabled
 - Azure DDoS Network Protection: Disabled
- IP addresses**
 - Address space: 10.0.0.0/16 (65536 addresses)

At the bottom, there are buttons for 'Previous', 'Next', and 'Create'. A 'Give feedback' link is also present in the bottom right corner.

The output appears as shown below:

The screenshot shows the 'myVnet | Overview' page in the Microsoft Azure portal. The page indicates that the deployment is complete. The following details are shown:

- Deployment name:** myVnet
- Subscription:** Simplilearn HOL 100758
- Resource group:** myRG1
- Start time:** 11/7/2023, 7:40:33 AM
- Correlation ID:** 6e9a6a93-bfe3-4a50-911a-a94500d16842

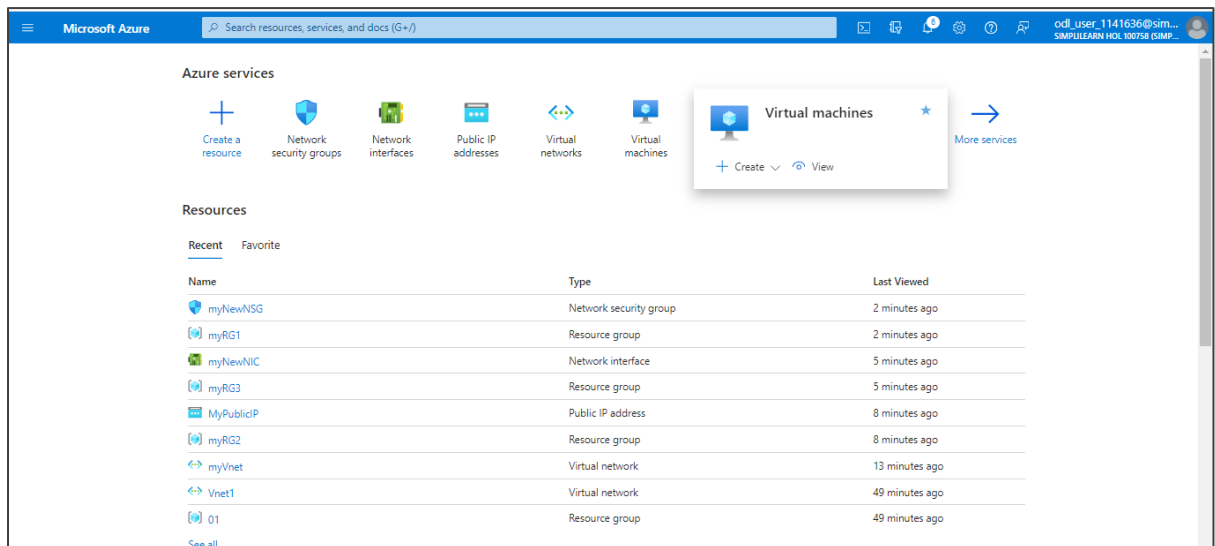
On the right side, there are three promotional cards:

- Cost management:** Get notified to stay within your budget and prevent unexpected charges on your bill. [Set up cost alerts >](#)
- Microsoft Defender for Cloud:** Secure your apps and infrastructure. [Go to Microsoft Defender for Cloud >](#)
- Free Microsoft tutorials:** Start learning today >

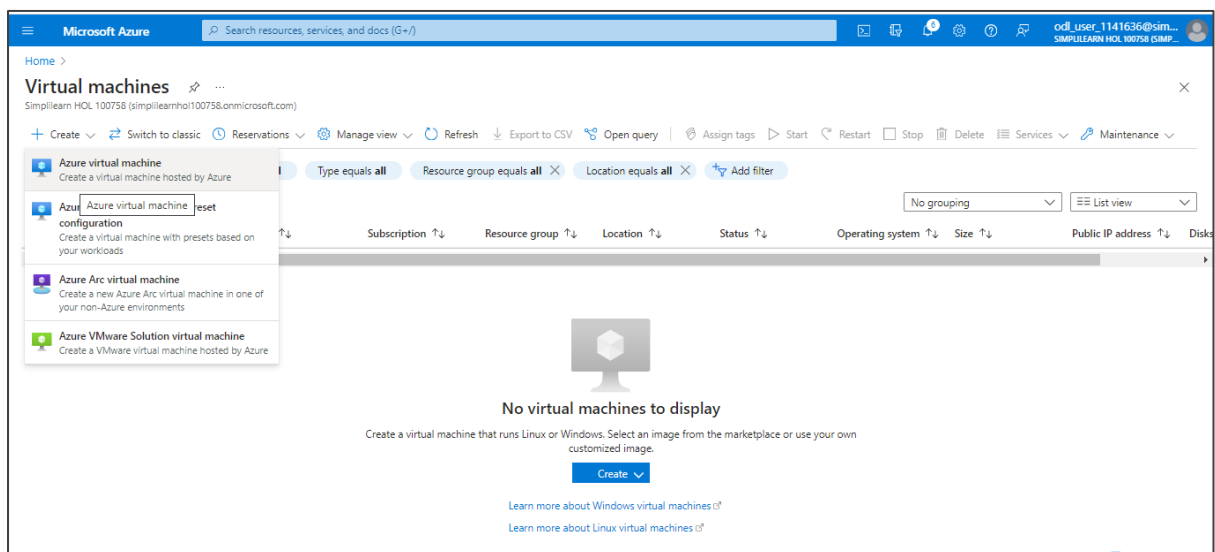
At the bottom, there is a 'Go to resource' button and a 'Give feedback' link.

Step 2: Deploy the virtual machine in the respective virtual network

3.1 In the Azure portal, search and select **Virtual machines**



3.2 Click on + **Create** and select **Azure virtual machine**



3.3 On the **Basics** page, enter the following details as shown below and click on **Next : Disks >**

Microsoft Azure | Search resources, services, and docs (G+)

Home > Virtual machines >

Create a virtual machine

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Instance details

Virtual machine name *

Region *

Availability options

Availability zone *

Microsoft Azure | Search resources, services, and docs (G+)

Home > Virtual machines >

Create a virtual machine

You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type

Image * [See all images](#) | [Configure VM generation](#)

This image is compatible with additional security features. [Click here to swap to the Trusted launch security type.](#)

VM architecture ☐ Arm64 ☒ x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount ☐

Size * [See all sizes](#)

Item(s) availability based on policy assignment(s) for the selected scope. az13049-PolicyDefinition ([Policy details](#))

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machines >

Create a virtual machine

Administrator account

Username * ✓

Password * ✓

Confirm password * ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ☐ None ☒ Allow selected ports

Select inbound ports *

i All traffic from the Internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

[Review + create](#) < Previous Next : Disks > [Give feedback](#)

3.4 On the **OS disk** page, under **OS disk type** select **Standard HDD (locally-redundant storage)** and click on **Review + create**

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machines >

Create a virtual machine

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host ☐

i Encryption at host is not registered for the selected subscription. [Learn more about enabling this feature](#)

OS disk

OS disk size

OS disk type *

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM ☒

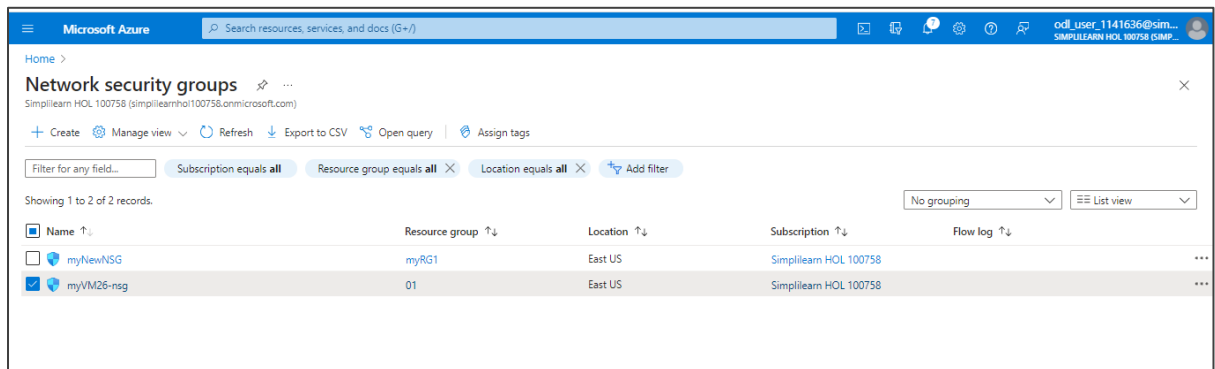
[Review + create](#) < Previous Next : Networking >

3.5 Click on the **Create**

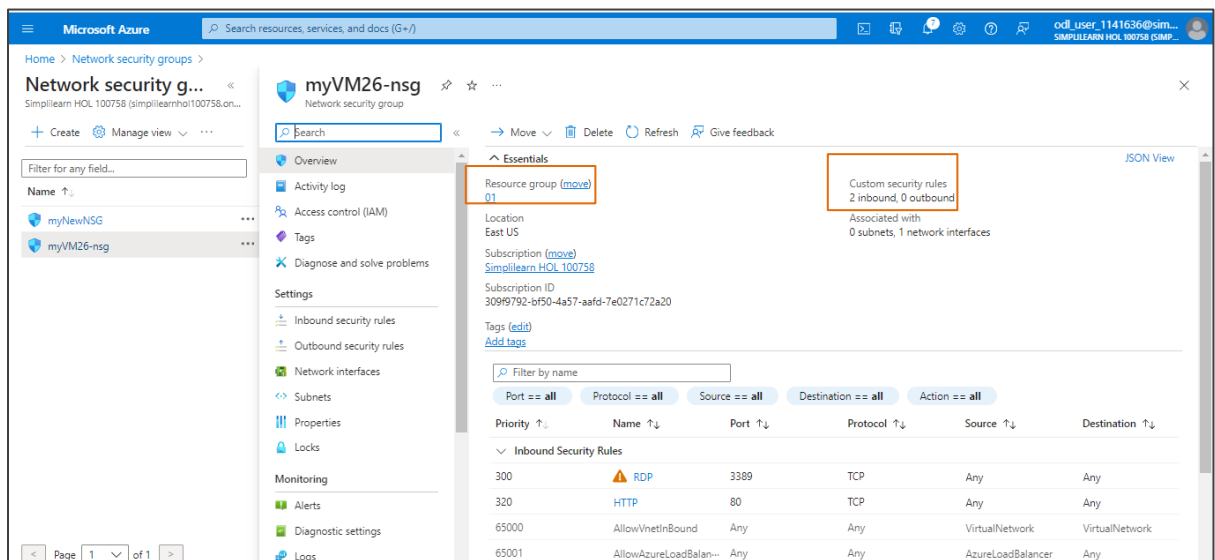
The output appears as shown below:

Step 4: Verify the network security group of the virtual machine

4.1 After the deployment of the virtual machine using the given virtual network under the respective resource group. The user can verify the NSG (network security group) of the virtual machine



4.2 The user can verify the network security group (NSG) associated with the VM. The NSG is crucial for controlling inbound and outbound traffic to the VM.



By following these steps, you have successfully created a network design that integrates on-premise infrastructure with Azure through a site-to-site connection, deploying multiple VMs.